

Desmos Mastery - Day 3

14 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Equivalent Expressions | 7 questions

[Lesson video: Equivalent Expressions](#)

1. Which expression is equivalent to the expression below for every real x ?

$$(2x - 1)(x + 2)$$

- A) $2x^2 + 3x - 2$
- B) $2x^2 - 3x - 2$
- C) $2x^2 + 5x - 2$
- D) $2x^2 - x + 2$

2. For $x \neq 3$, which expression is equivalent to the expression below?

$$\frac{x^2 - 9}{x - 3}$$

- A) $x + 3$
- B) $x - 3$
- C) $x^2 + 3$
- D) 1

3. Which expression equals the expression below for all real a and b ?

$$(a - b)^2 + 4ab$$

- A) $a^2 + b^2$
- B) $a^2 + 2ab + b^2$
- C) $a^2 + 3ab + b^2$
- D) $a^2 - ab + b^2$

4. The identity below holds for all real x . What is k ?

$$(x + 5)^2 = x^2 + kx + 25$$

Answer: _____

5. Which expression is equivalent to the expression below when $t > 0$?

$$\frac{(t^{3/2})^4}{t^2}$$

A) t^2

B) t^3

C) t^4

D) t^8

6. A student compares $(x + a)^2$ and $x^2 + a^2$ using $a = 0$ and sees matching graphs. Which statement is correct?

A) They are equivalent for every a .

B) Their difference is $2ax$, so the special test hides the difference.

C) They can never be equal.

D) Replacing x with a proves equivalence.

7. For $x \neq 0$, the expression below equals $ax + b$. What is $a + b$?

$$\frac{3x^2 + 4x}{x}$$

Answer: _____

Deeper Dive into Functions | 7 questions

[Lesson video: Deeper Dive into Functions](#)

8. For the functions below, what is $f(g(2))$?

$$f(x) = x^2 + 4, \quad g(x) = 2x - 1$$

Answer: _____

9. If $f(x) = 3x - 7$, what input t satisfies $f(t) = 20$?

Answer: _____

10. The graph of g is the graph of f shifted 4 units left and 2 units up. Which equation defines g ?

A) $g(x) = f(x - 4) + 2$

B) $g(x) = f(x + 4) + 2$

C) $g(x) = f(x + 2) + 4$

D) $g(x) = f(x + 4) - 2$

11. If $f(x) = x^2 - 2x$, what is $f(-3)$?

Answer: _____

12. If $f(x) = 2x + 5$ and $g(x) = x^2$, which expression equals $g(f(x))$?

A) $2x^2 + 5$

B) $4x^2 + 25$

C) $4x^2 + 20x + 25$

D) $2x^2 + 20x + 5$

13. The function below has zeros at -2 and 6 . What is $f(2)$?

$$f(x) = (x + 2)(x - 6)$$

Answer: _____

14. For the function below, what is the greatest value of f on its stated domain?

$$f(x) = 10 - (x - 4)^2, \quad 0 \leq x \leq 3$$

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: A 2: A 3: B 4: 10 5: C 6: B 7: 7 8: 13 9: 9 10: B 11: 15 12: C 13: -16 14: 9

Why each answer works

- 1. Answer: A.** Multiply each term: $2x^2 + 4x - x - 2 = 2x^2 + 3x - 2$.
- 2. Answer: A.** Factor the numerator as $(x - 3)(x + 3)$, then cancel $x - 3$ only on the permitted domain. The original expression remains undefined at $x = 3$.
- 3. Answer: B.** Expanding $(a - b)^2$ gives $a^2 - 2ab + b^2$; adding $4ab$ gives $a^2 + 2ab + b^2$.
- 4. Answer: 10.** Expand the left side to $x^2 + 10x + 25$. Match the x -coefficient, $k = 10$.
- 5. Answer: C.** Multiply exponents first: $(t^{3/2})^4 = t^6$. Divide by t^2 to obtain t^4 .
- 6. Answer: B.** The missing cross term is $2ax$. It vanishes when $a = 0$, so that parameter choice provides misleading evidence.
- 7. Answer: 7.** Divide each numerator term by x : $3x + 4$. Thus $a = 3$, $b = 4$, and $a + b = 7$.
- 8. Answer: 13.** First $g(2) = 3$. Then $f(3) = 3^2 + 4 = 13$. Reversing the composition gives a different result.
- 9. Answer: 9.** Solve $3t - 7 = 20$, so $t = 9$.
- 10. Answer: B.** Replacing x by $x + 4$ shifts the graph left 4; adding 2 shifts it up 2.
- 11. Answer: 15.** Substitute the entire input: $(-3)^2 - 2(-3) = 9 + 6 = 15$.
- 12. Answer: C.** g squares its complete input: $(2x + 5)^2 = 4x^2 + 20x + 25$.
- 13. Answer: -16.** At $x = 2$, the factors are 4 and -4 , so their product is -16 .
- 14. Answer: 9.** The unrestricted maximum is 10 at $x = 4$, outside the domain. On $0 \leq x \leq 3$ the function increases; $f(3) = 9$ is the greatest permitted value.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.